BRAIN BENCHMARK · TELEGRAM PLAYBOOK

Test the change.
Then decide what to ship.

A practical guide to changing brain parameters, code and formulas through **@phorvexbot**: write a precise request, inspect the tests, and compare your candidate with the baseline brain.

English edition | Copyable examples | Parameter and code workflows | HTML + CSV reports

Guide version: **2026-10-28** · Based on the current bot implementation

Jump to the /bench walkthrough →

01 What does the bot do?

Test a change before release using recorded production states.

You describe a change in Telegram. The bot compares the baseline brain with your candidate on a set of **recorded ticks** and returns report files. A tick is the recorded state of one brain decision cycle.

01 / REQUEST	02 / PATCH	03 / TEST	04 / REPLAY	05 / REPORT
Describe the change File, function and behavior	Create a candidate From the baseline commit	Test and build Baseline and candidate	Replay the same data Compare decisions	Review the evidence HTML, CSV, diff and tests

What does the result actually establish?

It shows which decisions the candidate would make on recorded past states. It does not measure live defense rates, actual SLO effects, or the future system trajectory after those new decisions.

These workflows do not apply the change to the live brain, merge the candidate, or deploy it. Live attacks and actuators belong to the separate live lab, which is not currently available through this bot.

02

Access and readiness

Set up access once, then run /check before an important experiment.

1

Open a private chat with the bot

Open @phorvexbot and press **Start**. If your account is not authorized yet, ask the bot owner for a fresh invitation, open it, and press Start. The owner can use `/invite` to create a one-use membership link valid for 24 hours.

Opening the public bot link does not grant membership. The bot does not reply to unauthorized accounts.

2

Get help, check dependencies, and download the playbook

Send /help and /check as separate messages. Use /guide whenever you want both English guide files attached in this chat:

First message · bot usage

```
/help
```

Second message · readiness check

```
/check
```

Download this English guide as HTML and PDF

```
/guide
```

3 Check the requirements for your workflow

Row in /check	Practical meaning
Telegram / allowlist / state	The bot, your authorization, and its job storage must be ready.
Dashboard / deployed brain-replay	The dashboard and verified replay identity are needed, along with ticks from the matching revision.
Operator repository	`/bench` needs the exact baseline brain commit in the server repository.
Free disk / Go / OpenAI	Needed to generate code, run tests and build a candidate. The default `/bench` free-disk threshold is 4 GiB.

A `WARN` can disable only one workflow. Read the row to see what is unavailable. An all-PASS readiness check does not guarantee that future tests or experiments will succeed.

03 Choose your workflow

Vary an existing override or change the decision logic itself.

`/param`

Compare values of one parameter

For example, try cost ceilings of 1.0, 1.2 and 1.5. The bot uses the deployed replay binary's override support. It does not generate or compile new code.

Parameter walkthrough →

`/bench`

Change code, a formula, or a condition

For example, increase the norm term's weight by 20%. A model generates a diff. The bot tests and builds the baseline and candidate, then compares them.

Code and formula walkthrough →

Topic	/param	/bench
Your input	Parameter ID and values	A precise description of a code change
Go tests / build	No; uses the deployed replay binary	Yes; baseline and candidate
Expected comparison status	`MATCHED_REPLAY`	`OBSERVATIONAL` with an `EXACT` baseline
Additional output	A supplementary Math Lab report	Candidate diff and test results

Shared command options

Option	Meaning	Important detail
`policy=vllm/vllm-defense`	Choose a namespace and policy	Use `namespace/name`. This is the current default policy.
`ticks=64`	Request up to 64 samples for an initial check	Current default: 500. Allowed request: 1–5000. Actual coverage depends on available, eligible data.
`nosync`	Use the data and replay identity already on the server	Without this flag, the bot syncs first. After a new deployment, omit it to refresh the identity and data.

> **With nosync, “current” means the last identity cached on the server.**
>
> The cache may lag behind a recent live deployment. Check the baseline commit in the report. Sync does not guarantee that the matching source commit exists in the operator repository; missing source must be fetched separately.

04

Change parameters with /param

Explore a supported override without editing code.

1

Choose the parameter and hypothesis

Example: “If I change the allowed cost ceiling, how do selected moves and estimated disruption cost change?” `P-FEA-006`, the **CHi cost ceiling**, is relevant to that question.

The ceiling is in normalized disruption units, not dollars. A lower value makes the cost gate stricter; a higher value makes it more permissive. This relationship alone does not establish better defense.

2

Run a small comparison

Example experiment · up to 64 cached ticks

```
/param P-FEA-006 1.0 1.2 1.5 policy=vllm/vllm-defense ticks=64 nosync
```

The bot automatically adds a control arm using the **recorded effective value** and replays each value you supplied as another arm. You do not need to enter the control value. Do not assume that 1.2 is the current value without checking the report.

Use ASCII digits and a decimal point. Each command accepts 1–11 experiment values. All numbers in this guide are examples, not recommendations for configuring the live brain.

3 Read the report, then increase the sample

Check status, actual sample size, and each value against the control. If the data and result are interpretable, try a larger sample. This example also syncs fresh data:

Larger sample with a fresh sync

```
/param P-FEA-006 1.0 1.2 1.5 policy=vllm/vllm-defense ticks=500
```

The sample or baseline revision may change between runs. Read every experiment against its own control. Check the cohort and commit before comparing numbers from separate reports.

Known parameters and their limits

ID	Meaning and unit	Domain and limitations
P-FEA-006	CHi ceiling; normalized disruption	Greater than 0, up to 1000. No `unset` override.
P-FEA-005	Proactive hygiene cadence; seconds	Whole seconds from 1 to 604800. The override changes both cadence and exposure-pressure scale. Fresh trusted event evidence can bypass the cadence gate.
P-OBJ-003	Fixed disruption aversion, or beta	0–10 and `unset`. Incompatible with `costPreferences`; preflight can return `MODE_CONFLICT`.

Second example · 5, 15 and 30 minutes, expressed in seconds

```
/param P-FEA-005 300 900 1800 policy=vllm/vllm-defense ticks=64 nosync
```

An ID in the registry does not mean every parameter can be overridden. The replay binary must support it and preflight must accept it. This cadence example is not a uniform throttle on every attack response.

05 Change code and formulas with /bench

Turn a clear hypothesis into a patch, tests, and a three-arm comparison.

> **/bench takes a description of a change.**
>
> In the current interface, you describe what to change and a model generates a diff. There is no direct branch or commit selector, uploaded-diff execution, or /rerun command.

1 Define one question you can evaluate

A useful example: “If I increase the norm term's weight in the cost penalty by 20%, does estimated disruption cost decrease? What happens to selected moves, predicted coverage, and no-op under attack?”

“Make the brain better” leaves the scope and evaluation unclear. Limit your first experiment to one function or condition. If you do not know the file or function, locate it in the source for the baseline revision first.

2 Include these five parts in the request

Part	Cost-penalty example
Location	`phorvex/internal/model/economics.go`, function `EconomicsMetadata.CostPenalty`
Mode	Only `independent-cost-preferences-v1`
New formula or behavior	Change the norm multiplier from 1 to 1.2; keep the flow term as it is.
Constraints	Preserve legacy and fixed-beta behavior, feasibility gates, SLO checks, and budgets.
Change size	Keep the patch minimal, without unrelated edits or broad refactoring.

Use exact file paths, function names and code identifiers. The technical English examples help retrieve relevant context. A fully Persian request without code identifiers may retrieve too little useful context. Verify the example's path and formula against your baseline commit.

3 State the old and new formulas explicitly

This is an example of defining an experiment. The suitability of 1.2 and whether the change passes existing tests have not been established.

Before · only in independent-cost-preferences-v1 mode

```
penalty = w_norm * norm + w_flow * flow
```

After · increase the norm contribution to the penalty by 20%

```
penalty = 1.2 * w_norm * norm + w_flow * flow
```

For arithmetic illustration only: with `w_norm=0.5`, `norm=2`, `w_flow=0.3` and `flow=1`, the penalty increases from 1.3 to 1.5. This describes the expression itself. The brain's move count or final estimated cost will not necessarily change by 20%.

4 Send the complete command as one message

Formula-change example · copyable experiment

```
/bench policy=vllm/vllm-defense ticks=64 nosync In phorvex/internal/model/economics.go, change EconomicsMetadata.CostPenalty in independent-cost-preferences-v1 mode from m.NormPreference.Value*norm + m.FlowPreference.Value*flow to 1.2*m.NormPreference.Value*norm + m.FlowPreference.Value*flow. Preserve legacy and fixed-beta behavior, feasibility gates, SLO checks and budgets. Keep the patch minimal.
```

> **Put options before the change description.**
>
> The order is /bench, then policy=..., ticks=... and optionally nosync, followed by the description. Options at the end can be interpreted as part of the model request instead.

To refresh data and the replay identity before the experiment, omit `nosync`. Use the command builder to change the sample size or other options.

5 Follow the execution stages

The bot gives you a job ID and reports progress. One job runs at a time; up to five can wait in the queue. Cold builds and tests can take several minutes. There is no fixed completion time.

Check the current job and queue

```
/status
```

Stage	What the bot does
Baseline and data selection	Checks replay identity and SHA256, selects ticks from the matching commit, and freezes the selected data for this experiment.
Patch generation	The model proposes a change against the baseline commit. The candidate is created in a temporary clone.
Tests	Runs Go tests on both versions: `internal/decision`, `cmd/brain-replay`, and changed packages. This is not the full repository CI suite.
Build	Builds baseline and candidate replays. Both use the same compiler release, read from the deployed binary's build metadata.
Replay and reporting	Replays three arms on the same ticks, then sends HTML, CSV, the diff and test results.

A failed test stops the job before the numerical benchmark.

If baseline or candidate tests fail, the bot sends the corresponding log. An intentional formula change can conflict with an existing numerical contract. Do not remove or bypass tests merely to get a benchmark result.

6 Keep all four outputs together

File	What to inspect
`report-<job>.html`	Status, commits, data coverage, fidelity, and comparison metrics
`metrics-<job>.csv`	Values, populations, intervals, data availability, and comparison type
`candidate-<job>.diff`	Whether the generated code implements exactly what you requested
`tests-<job>.json`	Baseline and candidate tests, packages, result counts, and toolchain release

File names include the job ID. Read the diff before interpreting the numbers. A successful build does not establish that the model understood your request correctly.

Every /bench request starts independently.

A second command does not continue the first candidate. It creates a fresh candidate from the baseline commit. Resending the same description with ticks=500 does not guarantee the same patch. Compare the diffs. Replaying a fixed candidate deterministically requires a separate engineering workflow.

First run: check the pipeline without changing behavior

To exercise patch generation, tests, builds and report delivery, request a comment-only change. This is a pipeline check, not an optimization. Conceptually, zero changed decisions is the expected result.

Pipeline check · add a comment only

```
/bench policy=vllm/vllm-defense ticks=64 nosync In phorvex/internal/decision/ablation.go, add one explanatory comment immediately before the existing if base.Ratio > 0 branch, explaining that the positive-ratio check avoids division by zero. Change comments only. Preserve all executable code and behavior.
```

Allowed candidate scope: Go files under `phorvex/`, excluding `phorvex/tools/` and `_test.go` files. Dependency, `go.mod`, chart and Dockerfile changes, and file deletions are not accepted by this workflow.

06

Build a copyable command

Choose your options, then send the output in Telegram.

This tool generates command text locally in this page. It does not connect to the bot or server. The example values are for learning, not production configuration recommendations.

The form checks ordering and some input constraints. File availability at the baseline commit, model output correctness, binary support, and preflight success are checked by the bot itself.

07

How to read the report

Establish comparison validity, examine metric changes, then assess your hypothesis.

1

Check status and actual data coverage

Report field	Meaning and next step
`COMPLETE`	The comparison finished with the expected coverage. This label alone does not establish improvement.
`PARTIAL`	Some data was ineligible or could not be compared. Read the denominators and exclusion reasons; do not treat this as equivalent to full coverage.
`FAILED` / `INVALID`	The result is not valid evidence for accepting the candidate. Resolve the reported problem first.
`64/64` or similar	Read actual counts separately for control and each arm. Requesting 64 or 500 ticks does not guarantee that many valid samples.

2

Understand the three /bench arms

`control`

The deployed replay binary

A tick is control-eligible only if this binary exactly reproduces the recorded production output.

base-local

The baseline source, built here

For a valid comparison, fidelity must be `EXACT` and its changed-decision share against control must be zero.

candidate

The code generated for your request

This is a different brain revision. Its results are labeled `OBSERVATIONAL`.

If base-local is DIVERGED, do not use the candidate result to make a decision.

First establish why the local baseline differs from the deployed binary. EXACT describes decision reproduction on the inspected ticks; it does not by itself guarantee that both binary files are byte-identical.

`/param` normally reports `MATCHED_REPLAY`. `OBSERVATIONAL` is expected for `/bench` because it compares different source revisions.

3 Read the metrics against your experiment question

ID	What it measures	How to interpret it
M-P21	Share of ticks whose selected action set differs from control	Behavioral sensitivity, with no inherent good/bad direction. It is not an improvement score.
M-P26	Selected moves per tick	A reduction may remove unnecessary moves or omit a needed response.
M-P28	Estimated disruption cost of selected moves	A cost-model estimate, not measured live disruption or money spent.
M-P19	Share of attack-backed asset-ticks with no selected move	An increase calls for inspecting no-op reasons and potentially missed responses.
M-P22	Share of attack-backed opportunities with complete evidence and an admitted positive-gain action	Describes a modeled opportunity, not a successful live defense.
M-P25	Predicted coverage using common control-arm weights	Useful for comparing modeled coverage; it is not an observed security outcome.

Metrics do not all have the same population or denominator. `n/a` means unavailable, not zero. Read intervals and sample counts alongside the headline values.

4 Separate changed behavior from improvement

A wholly hypothetical example for learning how to interpret results. These numbers do not come from a real bot run.

Metric	Control	Candidate	What this metric alone says
Changed decisions M-P21	0%	25%	One in four compared ticks has a different selected set.
Moves per tick M-P26	0.8	0.6	Mean selected moves are 25% lower.
Estimated cost M-P28	0.40	0.32	Mean modeled cost is 20% lower.
No-op under attack M-P19	10%	15%	Up 5 percentage points; inspect possible adverse effects.

This does not establish that “the brain improved by 25%.”

Estimated cost decreased, but no-op under attack also increased. Interpret the metrics together with the hypothesis and safety constraints. A single number cannot determine release acceptance.

In the CSV, `ARM_LEVEL` is the arm's value and `PAIRED_DIFF_VS_CONTROL` is a difference over matched pairs. Keep them distinct. `M-P18` uses each arm's own objective pricing, so comparing raw values after changing the cost formula can be misleading.

A recorded pipeline verification, with no optimization claim

In the recorded check on **2026-10-28**, job `0deacfd4c571` completed on baseline `0058a3edd2eb` with a comment-only candidate: 64/64 ticks, EXACT baseline fidelity, and 398 PASS test events, 2 SKIPs and zero failures per version. Changed decisions were zero. This is historical evidence that the pipeline ran. Use /check and a fresh report to assess the current state.

08 Troubleshooting

Keep the job ID, failed stage and error message together.

Symptom or message	Likely meaning	Next step
The bot does not reply	Your account is unauthorized, the invitation is used/expired, or the service is unavailable.	Try the private chat and ask the owner for a fresh invitation. If you are already authorized, the service needs inspection.
`BOT_REPO does not contain the deployed commit`	The new binary exists, but its exact source commit is missing from the server clone.	The operator should fetch that commit into BOT_REPO, then rerun `/check` and start a new experiment. A tag is not required.
`no recorded ticks` or revision mismatch	No matching-revision data is available for the chosen policy.	Check the policy and run without `nosync`. If recording is inactive, inspect that policy's capture setup.
Free disk below the threshold	The `/bench` start guard refused the job.	Free space on the server, then run `/check`. Lowering the guard alone does not solve a space shortage.
`MODE_CONFLICT`	The parameter conflicts with the current economics mode, such as beta with costPreferences.	Choose a compatible parameter. Do not change the live economics mode just to force this experiment.
Invalid value / unsupported parameter	The ID, domain, unit, or override support is wrong.	Use ASCII numbers, the right unit and a supported parameter. Read the preflight message.
`LLM step failed` or refused patch	There is a model-response, context, or file-scope problem.	Supply an exact file/function and a smaller change. Remove out-of-scope edits. Authentication/quota errors need service-configuration inspection.
`base-local Go tests FAILED`	Even the baseline failed tests on the server.	Inspect the baseline log. Code, tooling or resource pressure may be responsible. Do not attribute this directly to the candidate formula.
`candidate Go tests FAILED`	The candidate violated an existing test.	Read the log. The change may be wrong, or the intended contract may need independent code and test review. Do not bypass tests to obtain numbers.
`build failed`	Compilation, toolchain, module or resource problem	Inspect the failed build stage and log. There is no valid benchmark result yet.
`DIVERGED`	The rebuilt baseline does not reproduce control decisions.	Stop interpreting candidate comparisons and resolve baseline fidelity first.
`n/a` or incomplete coverage	Required evidence, a valid denominator or replayable samples are missing.	Read data_status, reason and sample counts. Do not record `n/a` as zero.

For troubleshooting, keep these items:

Job ID, the submitted command, the failed stage, and that job's log or report. Resolve the first failure before queuing several similar expensive experiments.

09 Review checklist

This supports a review. Completing it does not automatically authorize a release.

- [ ] I defined the hypothesis, target file or parameter, and evaluation metrics before running the experiment.
- [ ] The baseline commit, policy and data freshness suit my question. I checked the effect of `nosync`.
- [ ] For /bench, I read the diff and confirmed it implements exactly the requested change.
- [ ] Both versions' tests PASS and baseline fidelity is EXACT. I reviewed SKIPs and the test scope.
- [ ] I checked status, actual sample counts, exclusions and candidate coverage.
- [ ] I did not mistake changed decisions for improvement. I read cost, coverage and no-op together.
- [ ] I interpreted missing values and intervals correctly, and chose a sufficient, relevant sample.
- [ ] I retained reports, diff, tests and job ID, with a separate review and live-evaluation path for accepting the code.

0 of 8 items checked. Selections are not saved when you close the page.

A useful result note looks like this:

“On the specified baseline commit and N valid samples, the candidate with this diff passed the targeted tests. Baseline fidelity was EXACT. Metrics X and Y changed by these amounts; Z was unavailable or adverse. This is replay evidence. Acceptance still needs code review and appropriate live evaluation.”

10 Quick reference and FAQ

Send control commands as separate messages.

Command	Purpose
`/start` or `/help`	Show bot usage to an authorized account
`/check`	Check prerequisites and service access
`/guide`	Download this English playbook as HTML and PDF attachments in the same chat
`/bench [options] <description>`	Generate and test a code candidate; options precede the description
`/param <ID> <values...> [options]`	Compare parameter values with the recorded control
`/status`	Current execution stage and queue
`/runs`	The five most recent history records
`/cancel`	Request cancellation of the current job after its current step. Stopping may not be immediate.
`/invite`	Owner only, in private chat: a one-use member invitation valid for 24 hours

See recent runs	Cancel the current job after its current step
/runs	/cancel

▼ Can I say “run the previous candidate again with 500 ticks”?

Not through the current interface. Each /bench request is independent and the model regenerates a patch. Identical descriptions do not guarantee identical diffs. Compare the diff, baseline and data between runs. There is no /rerun command.

▼ Can I describe the change entirely in Persian?

You can write a Persian description, but include exact file paths, function names and English code identifiers. Context retrieval currently relies on ASCII identifiers. A fully Persian description may retrieve too little useful context. The technical English examples in this guide are a more reliable way to specify the request.

▼ Will the bot update old tests when a formula changes?

No. _test.go files are outside candidate scope. Establish whether the new code is wrong or the intended product contract is changing. A contract and test change needs separate review. The bot reports the failure log and stops before benchmarking.

▼ Do passing tests approve the entire brain and release?

No. These targeted tests do not cover the full CI suite, every package, or live tests. Review the diff, evidence quality, relevant metrics, and live effects in proportion to the change.

▼ Zero changed decisions: does that mean the formula has no effect?

It only means the selected action set did not differ on this experiment's valid ticks. The changed path may not have been active, another gate may have determined the result, rankings may have stayed the same, or the sample may be uninformative. Inspect the diff and experiment context first.

▼ How do I open the HTML report?

Download the report from Telegram and open it in a browser. Use the CSV in Excel or a data tool; read the diff in a text editor. This guide is also a self-contained file and can be read offline.

▼ Will repeated experiments with 1.2 always give the same result?

The data, baseline identity and selected sample can change between runs, especially after syncing. With /bench, the patch is also regenerated. Compare runs using their job IDs, commits, diffs and cohorts.

▼ What happens after I find a promising candidate?

The bot does not publish the change. Take the diff and evidence through code review, CI, and live evaluation appropriate to the change's risk. This stage gives you concrete evidence for that decision. Replay alone does not authorize applying the candidate to production.

Source material: `benchmark/bot/service.py`, `planner.py`, `candidate.py`, `replay_build.py`, `code_experiment.py` and `report.py`; the parameter registry and evidence in `benchmark/bot/VALIDATION.md`.

Prepared on **2026-10-28**. Check /help and actual /check results for later service changes. This file contains no passwords, tokens or private invitation links.